

RoadMap to Master Data Scientist Skills in 4 Months

Created by AI Coach John, CEO of ProITbridge.

Become a Data Scientist by Upskilling Yourself in Just 4–6 Months – Your Roadmap Starts Here.

This isn't just another roadmap. This is your personalized, no-fluff guide to upskilling and becoming a job-ready Data Scientist...fast!

Built especially for the fresh graduates of 2025 who are serious about landing their first IT job in the **next 5 to 8 months.**

Confused by machine learning? Not sure if Python or Statistics comes first? Feeling overwhelmed by the AI buzz? This is your GPS to clarity.

Over the next four months, we will focus on essential skills and alternate paths based on your strengths...whether you love building models with Python and Scikit-learn or exploring data with SQL, Excel, and Power BI.

Reminder: **You're not aiming for MAANG on Day 1.** You're building real-world skills, landing a solid first job, and growing your career from there.

Let's get started on this exciting journey!

Targeted Job Position

- Data Scientist
- ML engineer
- CV Engineer
- NLP Engineer
- AI Engineer
- Data Science Consultant
- Business Intelligence Engineer

Salary Range - 4 to 8 LPA

Step 1: IT Job Market Survey

99% of the Freshers will exclude this step and that's why they suffer

Step 1	Do research in IT Job Market(Recruitment Sites) and find if we have opportunities, skills expected, Salary offered, etc.,
Step 2	Get IT Job Market Awareness
Step 3	You can take informed decisions here
Step 4	Start with the Commitment and Own the Consequences - Both positive and negative during the process
Step 5	You can win with Consistency in the Plan

Month 1 – Python Foundations, Data Handling, SQL, Statistics, Regression & Git

Week 1: Python Setup, Core Programming & Pandas Basics

Topics:

- Python Installation & Environment Setup
- Explore IDEs: VS Code, Jupyter, Colab
- Jupyter Notebook Basics
- Python Basics: Syntax, Variables, Strings, Input/Output
- Loops, Functions, Conditional Statements
- Python Libraries Exploration: Pandas, NumPy, Seaborn, Plotly
- Python Data Structures: Lists, Tuples, Dictionaries, Sets

Week 2: Pandas Deep Dive, Data Cleaning & Visualization

Topics:

- DataFrames, Series, Indexing, Selection
- Data Input/Output
- Data Cleaning Techniques
- Handling Missing Data
- Data Visualization with Matplotlib, Seaborn, Plotly
- NumPy Arrays, Broadcasting, Indexing

Week 3: OOPS and Introduction to Statistics

Topics:

- OOPS Basics, Classes, objects, Constructors, Data Input/Output
- Inheritance Polymorphism, Encapsulation, Abstraction

- Statistics and Types, Sampling Techniques, Standard Deviation
- Five Number Summary, Random Variables, Probability distribution

Week 4: Hypothesis Testing

Topics:

- Inferential Statistics, Hypothesis Testing, Confidence Intervals
 - Population vs Sample, Sampling Methods
 - Types of Hypothesis Testing, Significance value and p-value
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Month 2 – SQL and Advanced Machine Learning

Week 5: Structured Query Language

Topics:

- Intro to SQL, Basic SQL syntax, data types, primary keys, and constraint
- Creating and Inserting into DB
- DQL and DML Commands and filtering data
- Aggregate Functions and Subqueries
- Group By and Joins, types of joins
- Window Functions and Stored Procedures

Week 6: Introduction to machine learning and algorithms

Topics:

- Multiple Linear Regression, Assumptions Check
- Evaluation Metrics, Cross-Validation, Hyperparameter Tuning
- Dataset Practice, Hands-on Coding, Feature Selection, Data Preprocessing
- Regularization (L1/L2), GridSearchCV, RandomizedSearchCV, Model Performance Tuning
- Logistic Regression, Confusion Matrix, Precision & Recall, ROC-AUC Decision Tree Theory, Decision Tree Script, Tree Pruning, GridSearchCV
- Overfitting in Trees, Bagging Theory, Random Forest Script, Model Validation Methods

Week 7: Model Optimization and Hyper-Parameter Tuning

Topics:

- Model Optimization Techniques, Gradient Boosting Regressor Theory
- Gradient Boosting Regressor Script, Gradient Boosting Classifier, XGBoost Theory & Script
- LightGBM Theory, Script, Feature Importance Plotting, Model Interpretability
- Hyperparameter Tuning

Week 8: Model Optimization and Hyper-parameter Tuning

Topics:

- KNearest Neighbour, Support Vector Machine
 - Hierarchical Clustering, KMeans Clustering, Cluster Validation Techniques, DBSCAN
 - Association Rules Theory, Association Rules Script, Recommendation Engine
 - Feature engineering, PCA, Dimensionality Reduction Techniques
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Month 3 – Deep Dive into Deep Learning , NLP and Deployment tools

Week 9: Introduction to Deep learning

Topics:

- Git and github, GIT commands ,Introduction to Deep Learning, , Real-World Applications
- Activation Functions Overview, Types and Activation Use Cases
- Optimizers (SGD, Adam, RMSprop), Loss Functions, MNIST Dataset Introduction, MNIST Classification

Week 10: ANN & CNN and architectures

Topics:

- ANN Hyperparameter Tuning Basics, Learning Rate Adjustment, Epochs & Batch Size Tuning, Regularization Techniques
- CNN & Concepts, LeNet-5 Architecture, AlexNet Architecture, CNN Script Implementation
- AlexNet on CIFAR-10, VGG16 & VGG19 Architectures, Transfer Learning Concepts, Inception Net V1/V2/V3

Week 11: Computer Vision and YOLO model

Topics:

- ResNet Theory, ResNet Architecture, Skip Connections
- OpenCV Basics, Image Processing with OpenCV, OpenCV2
- YOLO Overview, Architecture, YOLO Custom Model Creation
- Autoencoders Theory, RFM Segmentation, Customer Segmentation Use Cases

Week 12: Natural Language Processing and Deployment Tools

Topics:

- Spam Classifier with NLP, Word2Vec Theory & Implementation, BiLSTM
- BiLSTM vs LSTM, GloVe Embeddings, Attention Mechanisms

- Transformer Architecture & Use Cases, BERT
 - BERT Fine-Tuning, NER
 - Streamlit UI and Data Display & Visualization
 - FastAPI Overview, Request Handling & Learn how to build fast web APIs using decorators
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Phase 2 - APPLY [1 Month]

Project 1: ABC (Can't disclose)

Title: xxx (Can't disclose)

Skills Used: xxx (Can't disclose)

Extras: (Can't disclose)

Project 2: ABC (Can't disclose)

Title: xxx (Can't disclose)

Skills Used: xxx (Can't disclose)

Extras: (Can't disclose)

Phase 3 - Placement Related [2 Months]

After Upskilling & Applying on Projects

Week 1:

- Build your Resume & Portfolio (GitHub, LinkedIn, Kaggle)
- Record short project walkthrough videos (YouTube or Loom)
- Identify top job portals (Naukri, LinkedIn, Instahyre, Hirect)

Week 2:

- 2 Mock Interviews (Technical + HR)
- Interview prep for:
 - Python, SQL Queries
 - Statistics & Probability
 - Scenario-based Business Questions

Week 3:

- Apply to jobs daily (target 10–15 quality applications/day)
- Attend webinars, workshops, AI bootcamps
- Network on LinkedIn, reach out to recruiters

Week 4 (and beyond):

- Track interview feedback and revise weak areas

- Learn from rejections & keep refining
 - Continuously keep working on projects
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Regards,

John-The AI Coach.

CEO & Founder of ProITbridge.

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Your journey starts today. Not tomorrow 💪 🔥